

A STUDY OF FINGERPRINT IMAGE FILTERING

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ABSTRACT

Fingerprint image enhancement is a crucial step in automatic fingerprint recognition. Quite a lot of approaches for filtering the fingerprint image have been suggested. Most of them perform oriented band pass filtering. However, such filters, for example, Gabor filters, may create spurious ridge structure information. This is harmful for the feature extraction and therefore is harmful for the automatic fingerprint recognition. This paper examines the properties of applying Gabor filter in the fingerprint image enhancement. It shows that the non-sinusoidal-shaped ridge structure, the ridge frequency estimation error and the small filter size are the causes of creating spurious ridge structures. As a solution, we suggest an adaptive oriented low pass filter instead of Gabor filter to avoid producing undesired harmful side effects for the automatic fingerprint recognition.

1. INTRODUCTION

A gray level fingerprint image can be considered as an oriented texture pattern that contains narrow ridges separated by narrow valleys. The uniqueness of a fingerprint for automatic verification is exclusively determined by the local ridge characteristics and their relationships. The two most prominent local ridge characteristics, called minutiae, are ridge ending and bifurcation. A critical step in fingerprint verification is automatically and reliably detecting these minutiae from fingerprint images. Unfortunately, fingerprint images obtained in practice are of poor quality due to the variations in impression conditions, skin conditions and acquisition devices. The ridge structures in such low quality fingerprint images are not always well defined. Therefore, the problem of filtering fingerprint images to facilitate minutiae detection is an area that has received significant research interest.

For filtering fingerprint images, various approaches have been suggested, such as using band pass filter [1], directional Fourier-transform filter [2] and fuzzy approach [3] to enhance or smooth the raw gray level fingerprint

image. O'Gorman and Nickerson in [4] and Mehtre in [5] performed image enhancement by using contextual/directional filters, or called matched filters. A. Jain *et al* in [6] and [7] detected ridges with two masks, which in principle perform some kind of band-pass filtering. L. Hong *et al* in [8] introduced a fingerprint image enhancement method that employed a Gabor filter.

Although different names and proposed by different authors, all approaches above actually perform low pass filtering along the ridge orientation and band-pass filtering along the orientation orthogonal to the ridge orientation. So these filters could be called oriented band pass filters. The oriented band pass filters undoubtedly increase the clarity of ridge structure or the contrast between the ridges and valleys and therefore gives a good visual appearance of the image. Moreover, it is also able to separate some linked ridges. However, it is reported that a fingerprint enhancement algorithm based on the oriented band pass filter may result in spurious ridge structures [8]. The spurious ridge structures are harmful for the automatic fingerprint verification since they introduce spurious minutiae. It is therefore worth studying the causes of the spurious ridge structure and whether and how the spurious ridge structure can be avoided.

2. EXAMINING GABOR FILTER IN ONE DIMENSION

The even-symmetric Gabor filter has the general form [8]

$$h(x, y; \phi, f) = \exp \left\{ -\frac{1}{2} \left[\frac{x_\phi^2}{\delta_x^2} + \frac{y_\phi^2}{\delta_y^2} \right] \right\} \cos(2\pi f x_\phi) \quad (1)$$

$$x_\phi = x \cos \phi + y \sin \phi, \quad (2)$$

$$y_\phi = -x \sin \phi + y \cos \phi, \quad (3)$$

where ϕ is the orientation of the Gabor filter, f is the frequency selection orthogonal to the filter orientation, and δ_x and δ_y are the space constant of the Gaussian envelop along x_ϕ and y_ϕ axes, respectively. To apply Gabor filter to fingerprint images, the filter orientation ϕ is determined by the estimated fingerprint local ridge orientation and the frequency characteristic of the filter, f , is determined by the estimated fingerprint local ridge frequency.

If we decompose Gabor filter into two orthogonal 1-D filters $hl(y_\phi)$ and $hh(x_\phi)$, one along the orientation ϕ and the other orthogonal to the orientation ϕ , we have

$$h(x, y; \phi, f) = hl(y_\phi) \cdot hh(x_\phi) \quad (4)$$

$$\text{with } hl(y_\phi) = \exp\left\{-\frac{y_\phi^2}{2\delta_y^2}\right\} \quad (5)$$

$$\text{and } hh(x_\phi) = \exp\left\{-\frac{x_\phi^2}{2\delta_x^2}\right\} \cos(2\pi f x_\phi) \quad (6)$$

where y_ϕ is the projected axis along the ridge orientation and x_ϕ is the projected axis orthogonal to the ridge orientation. It is straightforward to see that Gabor filter performs a low pass filtering $hl(y_\phi)$ along the ridge orientation and a band pass filtering $hh(x_\phi)$ orthogonal to the ridge orientation. If the parallel ridges and valleys in a fingerprint image consist of ideal sinusoidal-shaped waves plus white noise, i.e. the 1-D signal orthogonal to the ridge orientation is a sinusoidal signal, Gabor filter that is tuned to the corresponding local ridge frequency and orientation can efficiently remove the undesired noise and preserve the true ridge and valley structure.

Unfortunately, many real fingerprints do not consist of ideal sinusoidal-shaped waves plus white noise. Fig. 1(a) shows a partial fingerprint image and an oriented window. Fig. 1(b) illustrates an *x-signature* of the image, i.e. 1-D signal orthogonal to the ridge orientation after 1-D filtering the image using equation (5) with $\delta_y=4$ and filter length of 11. Fig. 1(c) shows the power spectrum of the *x-signature*. We see from Fig. 1(b) that the *x-signature* of ridges and valleys is not a sinusoidal-shape signal. For a non-sinusoidal period signal, its power is distributed to its fundamental frequency and harmonics. In Fig. 1(c), the fundamental frequency and the second and third harmonics can be clear identified.

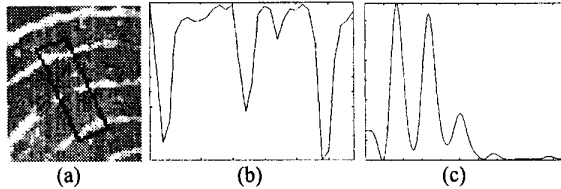


Fig. 1. A partial fingerprint image and an oriented window (a), *X-signature* (b) and its power spectrum (c).

The 1-D signal orthogonal to the ridge orientation is often a rectangular pulse train adding noise, which has not only fundamental frequency component but also many harmonics. For such signal, band pass filter may create spurious ridge structures if the filter frequency characteristic, f , deviates from the ridge frequency or the filter window size is small relative to the ridge period. The exact estimation of the ridge frequency is a difficult task, especially in the noisy regions and in the regions in which

the minutiae or singular points exist. For a 500dpi fingerprint image, the ridge frequency has range of $[1/3, 1/25]$. This large variance makes it difficult to choose an appropriate filter window size.

In [8], the filter window size and Gaussian space parameter δ_x were chosen to be 11 and 4, respectively. To explore the filter characteristics of equation (6) we keep these two parameter same as in [8]. To compare Gabor filter with the low pass filter, we choose the Gaussian envelop of equation (6) as our low pass filter, i.e.

$$hc(x_\phi) = \exp\left\{-\frac{x_\phi^2}{2\delta_x^2}\right\} \quad (7)$$

The cut off frequency of this low pass filter is controlled by adaptively adjusting the filter window size. We choose the filter size to be $2/(3f)$, where f is the estimated ridge frequency. Fig. 2(a) illustrates a rectangular pulse train of period 8 to simulate the *x-signature* of ridges and valleys with ridge and valley widths of 6 and 2 respectively, where two ridges are linked. Suppose there is a ridge frequency estimation error of 3 pixels, i.e. the filter is designed using $f=1/(8-3)$. Fig. 2(b) shows the output of 1-D Gabor filter of equation (6) and Fig. 2(c) shows the output of low pass filter of equation (7). It is clear seen that Gabor filter creates spurious ridge structures but the low pass filter not.

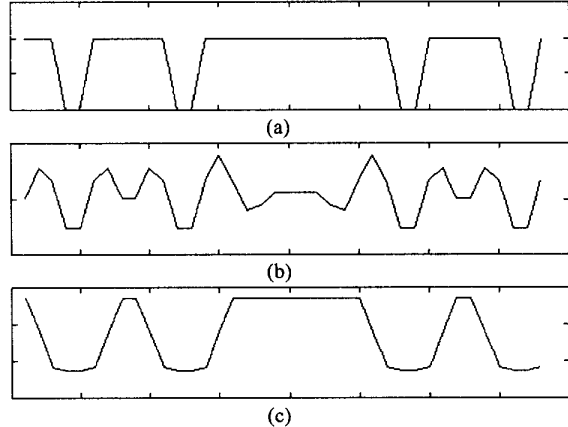


Fig. 2. A simulated *X-signature* (a), the output of 1-D Gabor filter with $\delta_x=4$ (b) and output of low pass filter with $\delta_x=4$ (c).

Changing the Gaussian space parameter δ_x cannot solve the ridge splitting problem of the Gabor filter. The four figures in Fig. 3 show the outputs of 1-D Gabor filter and low pass filter with the Gaussian space parameter δ_x of 8 and 2, respectively. From Fig. 3 (b) we see the degree of the ridge splitting is decreased with the small value of δ_x . However, the ability of separating two linked ridges is also decreased at the same time.

Even if there is no error of the ridge frequency estimation, spurious ridge structure may also be created by Gabor filter if the ridge period is larger than the filter size.

Fig. 4(a) illustrates a rectangular pulse train of period 14 to simulate the *x-signature* of ridges and valleys with ridge and valley widths of 12 and 2 respectively, where two ridges are linked. Gabor filter is designed using the exact ridge frequency, i.e. $f=1/14$. Fig. 4(b) shows the output of 1-D Gabor filter and Fig. 4(c) shows the output of the low pass filter, both with $\delta_x=4$.

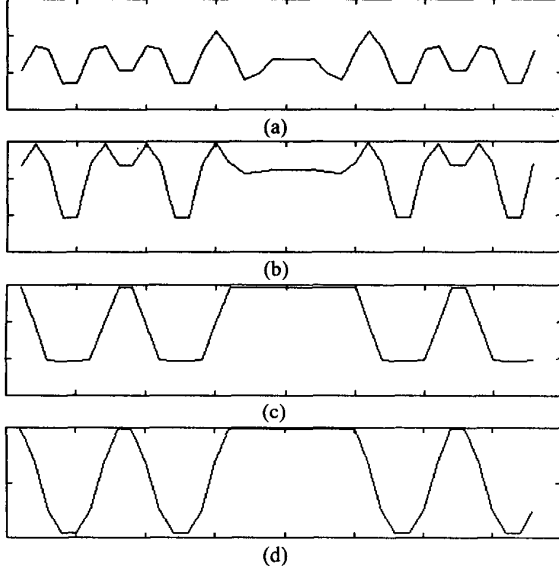


Fig. 3. Outputs of 1-D Gabor filter with $\delta_x=8$ (a) and $\delta_x=2$ (b) and outputs of low pass filter with $\delta_x=8$ (c) and $\delta_x=2$ (d).

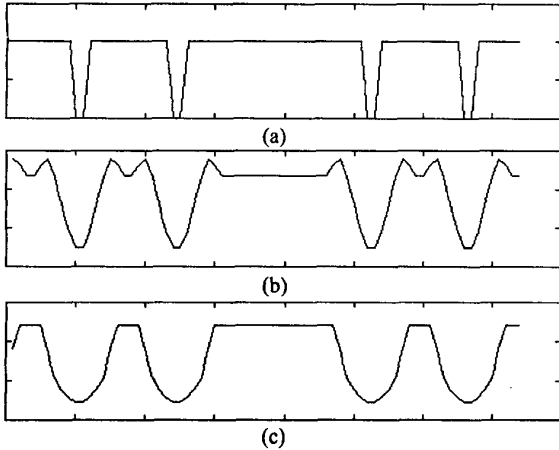


Fig. 4. A simulated *X-signature* (a), the output of 1-D Gabor filter with $\delta_x=4$ (b) and output of low pass filter with $\delta_x=4$ (c).

Outputs of 1-D Gabor filter and low pass filter with larger and smaller Gaussian space parameter δ_x are depicted in Fig. 5. Although Gabor filter with $\delta_x=2$ creates almost no spurious ridge structure, the linked ridge also cannot be separated (see Fig. 5 (b)).

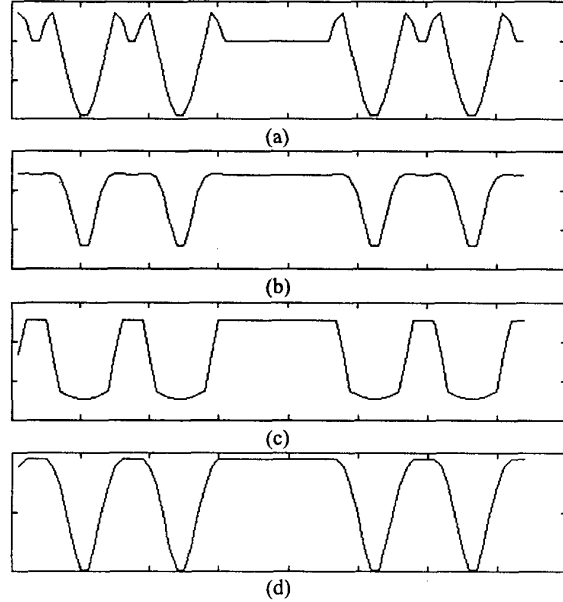


Fig. 5. Outputs of 1-D Gabor filter with $\delta_x=8$ (a) and $\delta_x=2$ (b) and outputs of the low pass filter with $\delta_x=8$ (c) and $\delta_x=2$ (d).

3. FINGERPRINT FILTERING SAMPLE RESULT

In this section, we apply Gabor filter of equation (1) and the suggested adaptive directional low pass filter (the product of the equation (5) and (7)) in the real fingerprint image. The filter size is chosen as 11×11 for Gabor filter and $11 \times [2/(3f)]$ for the adaptive directional low pass filter, where f is the estimated ridge frequency. The values of δ_x and δ_y are both set to 4. The local ridge orientation and frequency are estimated by using the method in [9] and [10], respectively. After filtering the image, a skeleton image and minutiae are extracted by using the algorithm described in [11].

A fingerprint image filtering approach is not aimed at producing a good visual appearance of the image but aimed at facilitating the subsequent ridge detection and minutiae extraction and avoiding undesired side effects in the subsequent processing. Therefore, we give the results of the ridge detection and minutiae extraction instead of the enhanced gray level images to compare the performance of different filters.

The white lines and black circles in Fig. 6 and Fig. 7 illustrate the skeleton image and minutiae extracted from the fingerprint image after employment of Gabor filter and the suggested adaptive oriented low pass filter, respectively. Some spurious ridge structures and the corresponding spurious minutiae can be seen in Fig. 6. From Fig. 7 it is clear that the problem of the spurious ridge structure does not occur and the location error of skeleton ridge is much smaller than that of Fig. 6.



Fig. 6. Extracted skeleton image and minutiae after using Gabor filter.



Fig. 7. Extracted skeleton image and minutiae after using the suggested adaptive oriented low pass filter.

4. CONCLUSION

Fingerprint image consists of oriented narrow ridges separated by narrow valleys. The local ridge orientation and ridge frequency are two intrinsic features of a fingerprint image. These two intrinsic features are used to design filters for noise attenuation. Oriented band pass filters, for example, Gabor filters, are widely used approaches in the literature to enhance the fingerprint image due to their frequency-selective and orientation-selective properties. This paper studied the characteristics

of Gabor filters when applying it in the fingerprint images. As we have seen from this study, since many real fingerprints do not consist of ideal sinusoidal-shaped waves plus white noise, ridge frequency estimation error and small filter size will create spurious ridge structures, which are harmful for the automatic fingerprint recognition. Due to the noise and bad image quality, the ridge frequency estimation error is unavoidable. Due to the curvature of ridges the large variance of the ridge frequency, the filter size should be limited and cannot be set large enough. In these cases, it showed that the separation of linked ridges of Gabor filter seems always to be accompanied with creating spurious ridges. As a conclusion, the application of Gabor filter in the fingerprint image enhancement seems problematic.

As this study showed, the adaptive oriented low pass filter suggested in this paper does not introduce any spurious ridge structure. Although a 1-D low pass filter orthogonal to the ridge orientation cannot separate linked ridges, a 1-D low pass filter along the ridge orientation may separate two ridges that are only shortly linked along the ridge orientation.

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